

CS-GY 6313: Information Visualization — Fall 2025

! This is a preliminary syllabus and subject to change !

Instructor: Robert Krueger (rk4815@nyu.edu)

Course Assistants : Erin McGowan (erin.mcgowan@nyu.edu), Yuancheng (Remo) Shen (ys6345@nyu.edu), Anushka Tawte (at5849@nyu.edu).

Class Information

Dates: 1/21/2025 - 5/6/2025

Meeting Times: Mondays 11am-1:30pm

Classroom: Tandon, Jacobs Academic Bldg Room 674 Loc: Brooklyn Campus

Discord: (will be added)

Course Prerequisites

The coursework includes extensive programming with JavaScript, the D3.js library, and web technologies (CSS, SVG, etc.). While previous knowledge of these technologies is not required, being proficient and comfortable with extensive programming is a fundamental prerequisite for this course. If you are not comfortable with programming please contact the instructor before enrolling.

Course Description

Being able to analyze and present data visually has become one of the most important skills for students who want to work in data science and related fields. Information Visualization teaches you how to design effective interactive visualizations of complex data for data understanding, discovery, and presentation.

The course is a blend of theoretical knowledge and practical work aimed at developing a well-rounded set of skills to ideate, design, implement, and evaluate sophisticated data visualizations. The theoretical part aims at providing a mental model to think about the visualization design space in a principled manner. This includes the theory of visual encoding, human perception and visualization techniques. The practical part aims at teaching the skills needed to develop effective interactive data visualizations for analysis and presentation. This includes teaching D3.js and JavaScript and practical labs on exploratory data analysis, sketching, graph design, analysis and critique.

The course also includes a series of small practical projects which enable students to gain experience with the development of fully-working interactive visualizations to solve non-trivial assigned problems. The work is organized in a way to simulate conditions happening in real-world data analysis and communication projects and includes activities to gain feedback from the instructor and the teaching assistant.

Workload

The course is intense. We assign multiple assignments every week (see details below) and we do a lot of practical work during the class. It is very important for you to do all the work on time otherwise you may get lost and catching up is hard. The good news is that you are going to learn a lot! Most of our students are extremely happy about how much they have learned by the end of the course. You are going to acquire very solid intellectual and practical skills.

Topics and Course Objectives

The lecture will cover the following topics:

What is Visualization / Why do we need it / Vis Pipeline, Data Questions, Task and Data Abstraction, Data Transformation, Visualization Types, Graphical Encoding, Marks & Channels, Perception, Color, Gestalt Laws and Principles, The Design Process (Sketch, Decide, Prototype, Evaluate), Temporal Vis, Spatial/Geo Vis, Network Vis, Multiple View and Interactivity. Depending on guest lecturer availability additional topics can include Scientific Visualization, Visual Analytics, AR/VR and Immersion, and Evaluation.

After this course, you should be able to:

- Describe what problems visualization can solve and how.
- Describe the process needed to design effective visualizations.
- Explain how to evaluate a visualization.
- Use appropriate graphs for a data analysis/presentation problem.
- Describe any visualization in terms of its individual components ("deconstruction").
- Identify and solve effectiveness problems with a given visualization.
- Generate alternative solutions for a data visualization problem (design space exploration).
- Assess pros and cons of alternative solutions (comparative analysis).
- Describe visual perception mechanisms that impact visualization effectiveness.
- Make design decisions grounded on principles of visual perception.
- Design/develop effective visualizations for temporal, geographical, network data.
- Design/develop effective interactive functionalities for a data visualization.
- Design/develop interactive multiple views for data visualization (dashboards).
- Use JavaScript to manipulate and transform your data.
- Use JavaScript and D3 to develop and deploy interactive visualizations on the web.

Study Material and Textbooks

- Lecture slides: Will be made available via Brightspace after each class.
- Research papers, articles and book excerpts: You can find them in NYU Brightspace in the Lessons tool.
- Observable notebooks: You can find them in our Observable collection of notebooks we have for the class.

The material will be available through Brightspace and a Google Drive folder we use for the course. You will find a link to the gdrive in Brightspace. The gdrive folder contains the following important sub-folders:

- In-Class Activities: The folder we use during the class (lab part) to share work done in groups.
- Lecture Slides: The slides presented in class will be online after each class.
- Q&A: The answers to the questions you pose every week in the quizzes regarding the last lecture.

The following books will be used for pre-readings but are also recommended in general. (For NYU students these are freely available via NYU library (digital)).

- Visualization Analysis and Design, Tamara Munzner, CRC Press 2014.
- Visual Thinking for Information Design, Colin Ware (The Morgan Kaufmann Series in Interactive Technologies) 2nd Edition
- Interactive Data Visualization for the Web, Scott Murray, O'Reilly Media 2017.

Course Organization

The course is designed as an in-class lecture (with participation elements like quizzes, discussions, sketching exercises) followed by an in-class lab with tutorials and programming exercises.

Exercises may be started in the lab but are designed to be continued as homework. The course also includes mini-projects and a group project.

The lecture will range between 1 hour and 1.5 hours, followed by a short break. The remaining (lab) time will be used for the tutorials and exercises.

Course Requirements

The course requires attendance, active participation and submission of all homework, including: exercises, quizzes and mini-projects. Students also have to complete a group project to be developed in teams of 3 students.

Exercises

The second half of the course (the lab) contains a set of exercises that can be started in the lab and continued at home as homework. The exercises are meant to teach you data visualization design/evaluation skills as well as the programming skills necessary to develop the solutions you design. The exercises count for 30% of your final grade. Each exercise can be resubmitted after grading. A re-submission can give you up to 50% of the lost points back, provided it addresses all the issues raised.

Topics of these exercises include:

- From questions to data transformations: create effective data questions and transformations.
- Chart design: use the appropriate chart for a given problem.
- Graphical decoding: deconstruct and critique visualizations in terms of their marks, channels, and gestalt principles.
- Graphical encoding: create many solutions for the same problem using different marks and channels.
- Color use and design: use the right coloring strategy for a different data visualization problems.

Quizzes

Each week's lecture will have a corresponding quiz to test your knowledge. The main purpose of the quizzes is to test your knowledge after attending the lectures and doing the readings. Research has shown that answering questions (retrieval) is the best way to improve your memory and understanding. For this reason, we will keep releasing quizzes throughout the course so that you can consolidate your knowledge. For this same reason the quizzes will include both questions about the topics you studied for the current week as well as topics you studied in previous weeks. This is the best way to learn!

Mini-Projects

In a mini-project you are asked to develop a data visualization for a specific data set and a specific set of questions. The assignment requires designing a solution and implementing it in D3/JavaScript. There is a total of 3 mini-projects, which students must develop individually, that is, **Students are not allowed to collaborate in developing the solutions.** All students are, on purpose, assigned the same projects. This will enable you to learn from the submissions your colleagues submit as well as the many discussions we will have during our meetings. The projects will be easier at the beginning and will gradually increase in complexity as the course progresses.

- Mini-project 1 (Geo): develop visualizations that answer specific questions with geographical data.
- Mini-project 2 (Time): develop visualizations that answer specific questions with temporal data.
- Mini-project 3 (Networks): develop visualizations that answer specific questions with network data.

Each mini-project has one optional additional question and visualization to develop designed to give extra credits. Each question (one for each mini-project) will give you up to a total of 2 additional grade points to be added to your final grade.

Group-Projects

The course includes also a group-project to develop in a team of 3 students. The group project gives you the opportunity to work on collaboratively with other students and to work on a topic of your own choice. The project is developed following a number of milestones that will be communicated during the course. Each student has to contribute equally to the development of the project. The total grade weight of the project is 30%.

Office Hours

The course assistants will provide office hours for individual questions that you might have about the homework assignments or project progress.

Grading

The final grade in this course is calculated according to the following weights:
(this is a preliminary syllabus. The grading is subject to change)

Grading breakdown:

- Quizzes: 10%
- Exercises: 30%

- Mini-Projects: 30%
- Group-Projects: 30%

Note: **we do NOT have a final or a midterm exam in this course.** The final grade will be computed from the grades you receive in individual assignments.

The final letter grade is calculated using the following grading schema:

[the grading schema will be added here]

We have a number of very important submission policies. It is very important for you to familiarize yourself with these policies and to keep them in mind. I will assume you have read and understood these policies when grading your assignments.

Re-submission policy:

In this course it is always possible to resubmit the exercises and the mini-projects after receiving a grade and feedback from your instructor. With a re-submission you can obtain up to 50% of the points lost in your first submission. Re-submissions must be requested (by sending a message to the Grader) and must be submitted within 1 week from the date when the grade and feedback are released. Failure to resubmit on time disqualifies the student for gaining the lost points.

Late and no-submission policy

While we are pretty flexible on re-submissions and grading, we are strict with late and missing submissions. This is for two reasons: (1) We want to prevent you from falling behind. If you fall behind, it is going to be very hard to catch up in this course; (2) If you submit after the solutions have been released and discussed in class you would have an unfair advantage. For this reason we have find a way to penalize late submissions (no hard feelings). Keep in mind: it is always preferable to submit something that is not optimal that not submit at all!

These are the rules we will apply:

Late -- Penalty

1-2 days -- 10% reduction of assignment's grade

3-7 days -- 25% reduction of assignment's grade

7 days -- 50% reduction of assignment's grade

no-submission -- [to be announced]

Exception: There s no re-submission and late-submission for quizzes.

Important: it is your responsibility to check that you understand what it is to be submitted. When an assignment is posted make sure to read it carefully and ask for clarifications as soon as possible. If you ask at the very last minute we may not be able to give you the feedback that you need on time! Similarly, it is your responsibility to understand when a given assignment is due. It is never a justification to say: "I made a mistake and thought the submission was due at a different time".

Extra-points: The mini-projects contain optional exercises that can be used to gain extra-points.

Devices in Class

We will use smartphones and laptops throughout the quarter to facilitate activities and project work in-class. However, research and student feedback clearly shows that using devices on non-class related activities not only harms your own learning, but other students' learning as well. Therefore, we only allow device usage during activities that require devices. At all other times, you should not be using your device. We will help you remember this by announcing when to bring devices out and when to put them away.

Collaboration, Plagiarism, Cheating Policies

The work students submit for individual assignments and class projects must be their own original work. When ideas are borrowed from existing work it is necessary to provide citations and a clear statement that describes which part has been adopted and which is original. For graded homework students are NOT allowed to collaborate with their peers. The submitted homework must be produced and submitted individually.

ChatGPT

It is important that the written work required by the course is yours. While you can use ChatGPT for own studying purposes you should NOT use ChatGPT or other AI tools for code or text generation for your submissions. Use of these tools for graded exercises, quizzes, and projects is considered academic misconduct.

Academic Dishonesty

It's always annoying having to explain that copying work or cheating is not allowed. I like to totally trust each and everyone of you. But bad things happen and I have to warn you that academic dishonesty is a very serious thing and you might get in very serious trouble if you get caught cheating. Students caught in dishonest behavior will be reported to the school.

As an NYU community member, it is your responsibility to know your rights and responsibilities around academic misconduct. Please click [here](#) to review the NYU Tandon Student Code of Conduct. Any questions about how this policy relates to this class, please ask your professor.

The School of Engineering encourages academic excellence in an environment that promotes honesty, integrity, and fairness, and students at the School of Engineering are expected to exhibit those qualities in their academic work. It is through the process of submitting their own work and receiving honest feedback on that work that students may progress academically. Any act of academic dishonesty is seen as an attack upon the School and will not be tolerated. Furthermore, those who breach the School's rules on academic integrity will be sanctioned under this Policy. Students are responsible for familiarizing themselves with the School's Policy on Academic Misconduct.

Academic dishonesty may include misrepresentation, deception, dishonesty, or any act of falsification committed by a student to influence a grade or other academic evaluation. Academic dishonesty also includes intentionally damaging the academic work of others or assisting other students in acts of dishonesty. Common examples of academically dishonest behavior include, but are not limited to, the following:

- Cheating: intentionally using or attempting to use unauthorized notes, books, electronic media, or electronic communications in an exam; talking with fellow students or looking at another person's work during an exam; submitting work prepared in advance for an in-class examination; having someone take an exam for you or taking an exam for someone else; violating other rules governing the administration of examinations.
- Fabrication: including but not limited to, falsifying experimental data and/or citations.
- Plagiarism: intentionally or knowingly representing the words or ideas of another as one's own in any academic exercise; failure to attribute direct quotations, paraphrases, or borrowed facts or information.
- Unauthorized collaboration: working together on work meant to be done individually (this includes sharing your work with other students for any purpose).
- Duplicating work: presenting for grading the same work for more than one project or in more than one class, unless express and prior permission has been received from the course instructor(s) or research adviser involved.
- Forgery: altering any academic document, including, but not limited to, academic records, admissions materials, or medical excuses.

Absence

Weekly class attendance is mandatory. We will check attendances in class.

Throughout the course you can miss class 1x without need to notify. After that:

For missed classes, please find the NYU School of Engineering Policies and Procedures on Excused Absences policy [here](#). Students who miss class for any reason should [complete this form](#), through the Office of Student Advocacy. Any questions about what qualifies as an excused absence should email advocacy.tandonstudentlife@nyu.edu.

An absence can be excused if you have missed no more than 10 days of school. If an illness or special circumstance has caused you to miss more than two weeks of school, please refer to the section labeled Medical Leave of Absence.

Students may request special accommodations for an absence to be excused in the following cases:

- Medical reasons
- Death in immediate family
- Personal qualified emergencies (documentation must be provided)
- Religious Expression or Practice

Deanna Rayment, advocacy.tandonstudentlife@nyu.edu, is the Director of Student Advocacy & Compliance and her office handles excused absences and general connection to resources. The Office of Student Advocacy is located in 5 MTC, LC240C and they can assist you should it become necessary or if you have any questions.

Moses Center Statement of Disability

If you are a student with a disability who is requesting accommodations, please contact New York University's Moses Center for Students with Disabilities at 212-998-4980 or mosescsd@nyu.edu. You must be registered with CSD to receive accommodations. Information about the Moses Center can be found at www.nyu.edu/csd. The Moses Center is located at 726 Broadway on the 2nd floor.

! This is a preliminary syllabus and subject to change !